

Subject: Update on Project Development: Change of Database and Technology Stack

Dear Mr. Leonard,

I would like to inform you about some technical updates made during the course of the project.

Initially, the project was using **Supabase** as the main database platform. However, I encountered recurring issues with certain data entries being rejected or not stored correctly. After extensive testing and troubleshooting, I decided to migrate the database to **Neon**, a more reliable and high-performance **PostgreSQL-based** platform. Neon offers improved scalability, better query performance, and seamless integration with modern backend tools.

Alongside this database migration, I have also implemented **Prisma** as the project's ORM (Object Relational Mapper). Prisma simplifies database interaction, ensures type safety, and makes it easier to manage models, queries, and migrations within the TypeScript environment.

The project's codebase now fully uses **TypeScript**, replacing plain JavaScript to enhance the application's reliability and maintainability.

Some key benefits of this updated stack include:

1. **Type Safety:** TypeScript and Prisma work together to detect type-related errors during development, preventing common runtime issues.
2. **Improved Maintainability:** Structured typing and Prisma schema management make the codebase easier to read, debug, and scale.
3. **Better Developer Experience:** Features such as autocompletion, inline documentation, and intelligent refactoring improve productivity and consistency.
4. **Enhanced Scalability:** The modular architecture of TypeScript, combined with Prisma and Neon's robust infrastructure, ensures long-term scalability.
5. **Seamless Integration:** Prisma's native PostgreSQL support allows efficient communication with Neon while maintaining full JavaScript compatibility after TypeScript compilation.

These changes were made to enhance the project's reliability, maintainability, and overall development efficiency. The migration to Neon and Prisma ensures smoother data management, faster queries, and more stable performance.

Please let me know if you'd like a more detailed breakdown of the migration process or configuration updates.

Kind regards,

Stephanus Sylvanus